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HalLoc: Token-level Localization of Hallucinations for Vision Language Models

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Abstract

Hallucinations pose a significant challenge to the reliability of large vision-language models, making their detection essential for ensuring accuracy in critical applications. Current detection methods often rely on computationally intensive models, leading to high latency and resource demands. Their definitive outcomes also fail to account for real-world scenarios where the line between hallucinated and truthful information is unclear. To address these issues, we propose HalLoc, a dataset designed for efficient, probabilistic hallucination detection. It features 150K token-level annotated samples, including hallucination types, across Visual Question Answering (VQA), instruction-following, and image captioning tasks. This dataset facilitates the development of models that detect hallucinations with graded confidence, enabling more informed user interactions. Additionally, we introduce a baseline model trained on HalLoc, offering low-overhead, concurrent hallucination detection during generation. The model can be seamlessly integrated into existing VLMs, improving reliability while preserving efficiency. The prospect of a robust plug-and-play hallucination detection module opens new avenues for enhancing the trustworthiness of vision-language models in real-world applications. The HalLoc dataset and code are publicly available at: https://github.com/dbsltm/cvpr25_halloc.

1. Introduction

Vision-Language Models (VLMs) [20, 35, 37] harness the language generation capabilities of Large Language Models (LLMs) [1, 28], along with the visual understanding provided by vision foundation models [7, 32], to comprehend and describe visual content in natural language. Although VLMs have shown remarkable performance across various multimodal tasks, they are prone to producing descriptions that are inconsistent with the visual content, a phenomenon commonly referred to as *hallucination* [18, 20, 26, 29].

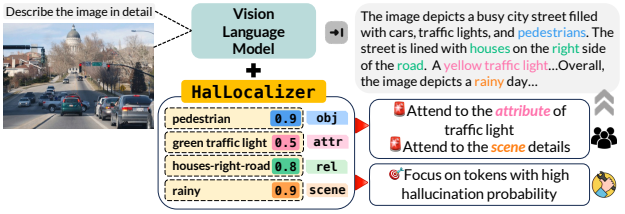


Figure 1. **Motivation behind HalLoc**: HalLoc aims to provide a practical, lightweight add-on for VLMs that simultaneously detects probabilistic hallucinations during response generation. By localizing tokens with high hallucination probabilities in real time, our model enhances use interaction by alerting at potentially unreliable content, prompting closer scrutiny or prioritizing external verification.

Previous studies [8, 18, 22, 25, 27, 31, 33] have explored the susceptibility of VLMs to various types of hallucinations, including those involving objects, visual attributes, spatial relations, and environmental details. The prevalence and diversity of these hallucinations pose significant challenges to real-world deployment, since they can lead to misinformation and undermine user trust [3].

Current approaches to hallucination detection often leverages powerful off-the-shelf models like GPT-4 [23] to cross-verify outputs [4, 13, 19]. While effective, these methods introduce substantial computational overhead and latency, bringing challenges to their applications requiring real-time feedback or operating under stringent computational constraints. Moreover, existing methods typically yield definitive outputs as either *hallucinated* or *truthful*. This dichotomy fails to capture the nuanced spectrum of confidence levels needed for ambiguous cases where the distinction is unclear.

Therefore, there is a strong need for an efficient hallucination detection mechanism that offers probabilistic assessments. A plug-and-play hallucination detection module, as illustrated in Figure 1, can be seamlessly integrated into existing VLMs. Such a system enables more nuanced interactions with model outputs, benefiting users and downstream applications. For instance, users can be alerted to content requiring additional scrutiny without dismissing it outright,

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Table 1. **Comparison of HalLocalizer** with existing hallucination detection methods. Note that *Discrete* output refers to definitive classes for detection and *Probability* output stands for probability as confidence levels of hallucinations.

Methods	Granularity	Type Classification	Detection	Output		VLM Tasks
				Discrete	Probability	
CHAIR [26]	Token	Object	Hand-crafted Rule	✓		Img Captioning
GAVIE [19]	Response	Object	GPT [23] as judge	✓		Instruction Following
FAITHScore [13]	Atomic Fact	None	off-the-shelf	✓		VQA, Img Captioning
UNIHD [4]	Segment	Obj, Attr, Scene-text, Fact	off-the-shelf	✓		VQA, Img Captioning
MHalDetect [9]	Sentence	None	Trained Model		✓	Img Captioning
***HalLocalizer* (ours)**	Token	Obj, Attr, Rel, Sce	Trained Model		✓	VQA, Instruction Following, Img Captioning

Table 2. **Comparison of HalLoc** with existing hallucination detection datasets.

Benchmarks	Size	Granularity	Type Annotation	VLM Tasks
HaELM [30]	56K	Response	None	Img Captioning
MHalDetect [9]	16K	Segment	None	Img Captioning
MHaluBench [4]	400	Segment	Obj, Attr, Scene-text, Fact	VQA, Img Captioning
***HalLoc* (ours)**	155K	Token	Obj, Attr Rel, Sce	VQA, Img Captioning, Instruction Following

and systems can prioritize external verification for content with a higher likelihood of being hallucinated.

To address this need, we introduce HalLoc, a large-scale dataset comprising 155K samples spanning visual question answering (VQA), instruction following, and image captioning tasks. Each sample is meticulously annotated at the token level with specific types of hallucinations, facilitating comprehensive training and evaluation of hallucination detection models.

Leveraging HalLoc, we develop a simple yet effective baseline model that performs concurrent hallucination detection with minimal computational overhead. Our model can be integrated into existing Vision-Language Models (VLMs) in a plug-and-play fashion, enhancing their reliability without sacrificing efficiency.

Our contributions can be outlined as follows:

1. Introduction of HalLoc dataset: We present HalLoc, a large-scale dataset comprising 155,000 samples annotated at the token level for specific types of hallucinations. Spanning tasks like visual question answering, instruction following, and image captioning, HalLoc facilitates comprehensive training and evaluation of hallucination detection models in VLMs.
2. Development of an efficient hallucination detection Model: Leveraging HalLoc, we develop a simple yet effective baseline model that performs concurrent hallucination detection with minimal computational overhead. Unlike existing methods that provide binary outputs, our model offers probabilistic assessments, capturing nuanced confidence levels for ambiguous cases.
3. Plug-and-Play integration with existing VLMs: Our hallucination detection model is designed to be seamlessly integrated into existing VLMs in a plug-and-play fashion. This enhances the reliability of VLMs without sacrificing efficiency, addressing the need for an efficient hallucination detection mechanism suitable for real-world applications with computational constraints.

2. Related Work

Hallucination Detection Methods Early work on hallucination detection focused on identifying non-existent objects in generated outputs. CHAIR [26] detects hallucinated objects by comparing mentioned objects with ground-truth data, using a simple rule-based approach limited to object-level hallucinations. Later approaches use powerful off-the-shelf models for detecting hallucinations. GAVIE [19] employs GPT-4 [23] to assign single accuracy scores to responses. FAITHScore [13] uses a chain of large models to break down responses into atomic facts and verify them via visual entailment. UNIHD [4] identifies and classifies hallucinations at the semantic segment level by also leveraging a chain of off-the-shelf models including GPT-4 [23].

HaELM [30] and MHalDetect [9] train models on specialized datasets to detect hallucinations at the sentence or response level. However, these approaches lack granularity as they detect hallucinations on long caption segments (inclusive of both hallucinated and non-hallucinated tokens) and do not provide hallucination type annotations, limiting their versatility.

As shown in Table 1, unlike previous approaches, our *HalLocalizer* trained on *HalLoc* identifies object, attribute, relationship, and scene hallucinations at the token level across diverse tasks. HalLoc's confidence-based detection allows precise maneuvering over ambiguous hallucination tokens. Also, its lightweight design allows for a seamless integration with VLMs.

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Hallucination Detection Datasets Previous works have crafted datasets to train automated hallucination detection models. HaELM [30] and MHalDetect [9] provide datasets with 56K and 16K samples, respectively, for image captioning with response-level or segment-level annotations. However, they do not verify over diverse tasks or include type-specific annotations, which are essential for a more comprehensive evaluation. MHaluBench [4] provides a broader evaluation framework featuring 400 annotated samples that span image captioning and VQA, with segment-level annotations for objects, attributes, scene-text, and fact hallucinations. However, its relatively small size makes it more suited for evaluation than large-scale model training.

As illustrated in Table 2, HalLoc distinguishes itself as the only dataset offering token-level granularity with annotations for objects, attributes, relationships, and scene types, supporting the broadest array of vision-language tasks. Despite the complexity of granular annotations, it remains the largest dataset with 155K samples, providing a strong foundation for training versatile automated hallucination detection models like HalLocalizer.

3. Hallucination Types and Patterns

3.1. Hallucination Types

Previous studies [8, 27, 31, 33] highlight that VLMs exhibit diverse forms of hallucinations, underscoring the need for comprehensive coverage. In *HalLoc*, we address four types of hallucinations caused by image-text misalignment: object, attribute, relation, and scene. To enable token-level detection, we provide a detailed taxonomy.

Object Hallucination (`<obj>`) A non-existent object. Only the token referring to the misidentified object constitutes an object hallucination. Attributes or relationships associated with the nonexistent object are excluded.

Attribute Hallucination (`<attr><obj>`) An incorrect property that is associated with a single object. Properties may be visual features, spatial positions, or actions involving a single object (e.g., *white shelf*, *door to the left of the image*, *sleeping woman*). Both the object token and the property token constitute an attribute hallucination.

Relationship Hallucination (`<obj1><rel><obj2>`) An unfaithful interaction involving two objects. Interactions include relative spatial positions or actions between two objects (e.g., *mirror above the sink*, *man holding an apple*). The two object tokens and the interaction token constitute a relationship hallucination.

Scene Hallucination (`<sce>`) An incorrect scene description of the image. Scenes include weather, place, or location (e.g., *sunny*, *office*, *outdoors*). Only the scene token constitutes a scene hallucination.

3.2. Hallucination Patterns

Accurate representation of hallucination patterns is vital to the quality of the dataset. Below, we outline common patterns guiding *HalLoc*'s generation pipeline.

Concept Association Bias Concept Association Bias (CAB) [34] occurs when vision-language models treat an input as a bag of concepts and attempt to infer missing concepts cross-modally. For example, if an image contains both a lemon and an eggplant, the model might incorrectly predict the lemon as purple, influenced by the strong semantic connection between *purple* and *eggplant*, rather than accurately relying on the visual cues.

Table 3. VQA results for probing statistical biases in hallucinations. We pose binary questions modeling VLMs to elicit hallucinations stemming from **Lang**uage or **Image** priors. The questions ask about attribute, relationship, scene-type hallucinations likely to co-occur due to these priors. **A**(ccuracy), **P**(recision), **R**(ecall), and **F1**(-score) metrics are reported. Yes % denotes the proportion of questions answered with "Yes".

Hallucination Type	Prior	A	P	R	F1	Yes%
Attribute	Lang	0.47	0.45	0.29	0.36	0.32
	Image	0.52	0.53	0.33	0.41	0.31
Relationship	Lang	0.71	0.69	0.75	0.72	0.53
	Image	0.64	0.61	0.74	0.67	0.60
Scene	Lang	0.35	0.41	0.66	0.50	0.81
	Image	0.33	0.39	0.63	0.48	0.80

Statistical Bias Hallucinations in VLMs have also been attributed to an overreliance on statistical bias [2] in unimodal priors [10, 11].
(1) *Language Prior*: The model generates a hallucinated response due to textual context alone, represented as $P(A_{\text{hal}} \mid Q) > P(A \mid Q)$.
(2) *Image Prior*: The model generates a hallucinated response due to the visual context, represented as $P(A_{\text{hal}} \mid I) > P(A \mid I)$.
(3) *Language-Image Prior*: The model generates a hallucinated response due to both the textual and visual context, represented as $P(A_{\text{hal}} \mid I, Q) > P(A \mid I, Q)$.

To demonstrate that VLMs exhibit these statistical biases, we pose binary (yes/no) questions targeting frequent attribute, relationship, and scene combinations. The binary questions are of the form: "Is `<obj> <wrong-attribute>`?" (e.g., *Is the lemon purple?*), "Is `<wrong-subj> <predicate> <obj>`?", "Is `<subj> <wrong-predicate> <obj>`?" The question set includes an equal number of questions with

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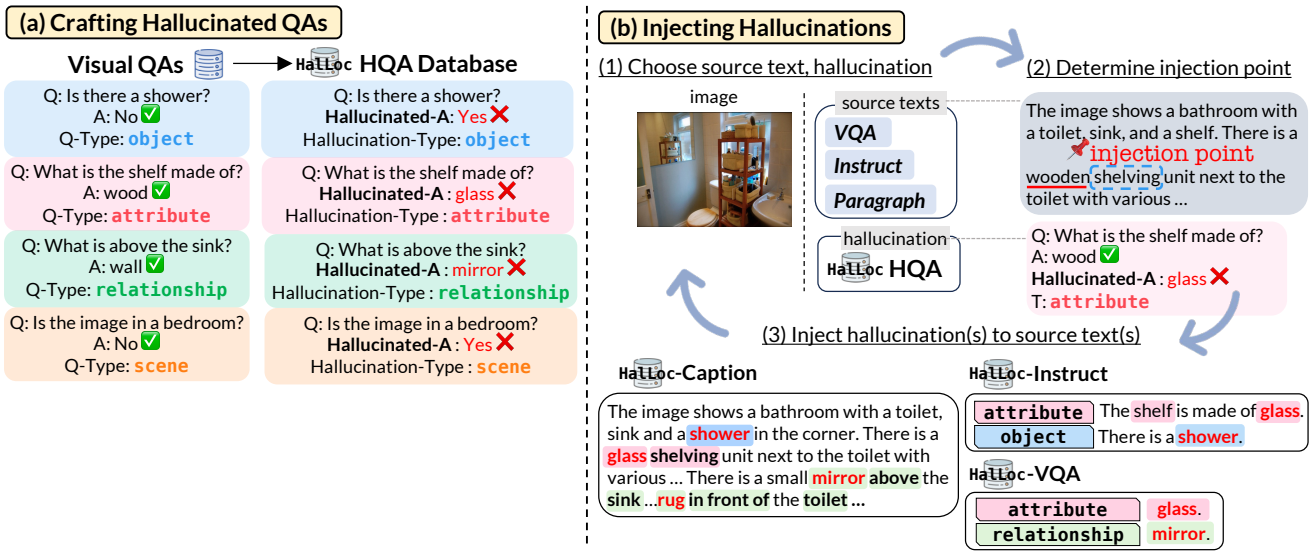


Figure 2. Overview of our automated hallucination annotation pipeline with HQA Injection. The pipeline consists of two stages: hallucination generation and injection. First, we (a) **craft** (question, hallucinated answer) pairs for each hallucination type, using strategies reflecting concept association biases statistical biases in VLMs (details are shown § 4.1). Then (b) GPT-4 [23] **injects** hallucinated QA pairs from the HQA Database into source texts that refer to the same image. Source texts span VQA, instructions, and paragraphs and vary for the three task types of *HalLoc* via text-based QA. Details for each source tasks description for each task is discussed in § 4.2.

"Yes" and "No" answers. VQA results in Table 3 show low performance in discerning highly co-occurring attributes and scenes. For scene hallucinations, the high "Yes" rate indicates that the model is overconfident, resulting in lower accuracy on questions where the correct answer is "No".

4. HalLoc

We present *HalLoc*, the first large-scale benchmark (155K samples) with token-level annotations of hallucination types spanning three core vision-language tasks: Visual Question Answering (VQA), Instruction-following, and Image Captioning.

4.1. Generation Pipeline

The comprehensive scale and fine-grained annotations of *HalLoc* are enabled by our *HQA-Injection Pipeline*, which allows for controllable and scalable hallucination annotation. The *HQA-Injection Pipeline* is a two-stage process: crafting hallucinated answers and injecting hallucinations. An overview of our *HQA-Injection Pipeline* is illustrated in Figure 2.

Crafting Hallucinated Answers We first construct a large Hallucinated Question-Answer (HQA) database, which contains questions, their corresponding hallucinated answers, and annotations of the hallucination types based on the taxonomy defined in Section 3.1. The questions are sourced from the GQA dataset [12], a Visual Question Answering benchmark featuring images from the Visual Genome dataset [16]. We select questions pertaining to the existence of objects, object attributes, relationships between objects, and environmental surroundings. Each question is labeled with its corresponding type (object, attribute, relationship, or scene).

Next, we employ two strategies to craft hallucinated answers for each question, aiming to mimic real-world hallucination patterns.

(1) We use GPT-4 [23] to select from hallucinated answer candidates originating from hallucination patterns observed in Vision-Language Models (VLMs), as defined in Section 3.2. For Concept Association Bias, we collect hallucinated answer candidates by sampling traits of the same category associated with different objects in the image. For example, in Figure 2 (a), the attribute hallucination *glass shelf* is created by borrowing the attribute *glass* from the *glass window* present in the image. For Statistical Bias, we gather data on the co-occurrence of objects, attributes, relationships, and scenes associated with a particular object. The candidates are subsequently categorized into specific types: language-prior, image-prior, or language-image-prior.

(2) We additionally use GQA [12] decoys or responses from VLMs. To capture hallucinations beyond the predefined types, we directly employ decoy answers or model-generated responses as hallucinated answers. Some GQA questions naturally have decoys (e.g., question: "Is the color of the shoe blue?" answer: "No" where color *blue* reflects concept association bias or co-occurrence biases), which

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we use as hallucinated answers. Furthermore, we conduct VQA on a subset of the selected questions using two popular VLMs: LLaVA 1.5 (7B) [20] and InstructBLIP (7B) [6]. After filtering out correct answers via exact match, a response is accepted as a hallucinated answer only if three strong VLMs unanimously agree that the response is not visually entailed. The three LVLMs used for verification are InternVL 2.0 [5] (26B), LLaVA 1.5 (13B), and InstructBLIP (13B).

Finally, following our defined taxonomy, we annotate the complete hallucination traits in each question-hallucinated answer pair. The type of the question naturally determines the hallucination type. The scene graphs provided in the GQA dataset allow for algorithmic annotations, which are subsequently verified by GPT-4 to eliminate any erroneous annotations.

Injecting Hallucinations With the large-scale HQA Database containing diverse types of hallucinated answers, we proceed to craft hallucinated responses for various tasks by systematically injecting these hallucinations.

First, we select source texts and questions from the HQA Database. The source texts are drawn from existing datasets designed for a variety of vision-language tasks and include prompts describing the task, accompanied by ground-truth responses free of hallucinations. We sample questions from the HQA database such that each source text and question shares the same reference image.

Next, we identify the injection point by querying the ground-truth response with the chosen question using GPT-4. If an answer is detected within the text, its position is designated as the injection point. If no answer is found, GPT-4 suggests a natural injection point within the text.

Finally, we inject the hallucinated answer into the text. If an answer was previously identified, it is replaced with the hallucinated answer. If no answer is found, GPT-4 generates a phrase or sentence containing the hallucinated answer and inserts it at the injection point. We annotate the complete component of the underlying hallucination trait. This process is challenging, even for a robust model such as GPT-4; therefore, we incorporate an additional verification step. For each question, the process is repeated up to three times until GPT-4 verifies that the injection is acceptable.

4.2. HalLoc Composition

HQA Database The HQA Database comprises 159,612 questions. Regarding hallucination types, there are 40,816 object-hallucinated answers, 75,627 attribute-hallucinated answers, 39,085 relationship-hallucinated answers, and 4,084 scene-hallucinated answers. The relative scarcity of scene hallucinations is attributable to the limited number of images with clearly discernible environmental settings, such as weather or location. The distribution of

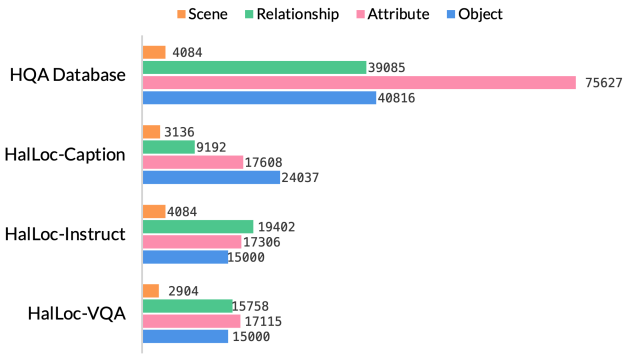


Figure 3. Number of data points per each hallucination type in the HQA Database, *HalLoc-VQA*, *HalLoc-Instruct*, *HalLoc-Caption*.

hallucination types across each subset of *HalLoc* is illustrated in Figure 3. In terms of hallucination patterns, there are 15,524 Concept Association Bias hallucinated answers, 42,671 language-image prior hallucinated answers, 16,546 language prior hallucinated answers, and 64,129 hallucinated answers derived directly from GQA decoys or incorrect VLM responses.

HalLoc HalLoc comprises 155,953 samples, divided into 124,528 training samples, 16,271 validation samples, and 15,154 test samples. Each sample contains an average of 17.66 words, with an average of 1.81 hallucinated words (about 10%). For comparative purposes, HalLoc also includes 13,713 non-hallucinatory samples. HalLoc is partitioned into three subsets: *HalLoc-VQA*, *HalLoc-Instruct*, and *HalLoc-Caption*.

HalLoc-VQA This subset contains 55,854 samples. The source text is derived from ground-truth answers in the GQA dataset. As it models the Visual Question Answering (VQA) task, the average number of words per sample is notably small—1.05 words, with an average of 0.96 hallucinated words (91%). It includes 15,000 object hallucinations, 17,115 attribute hallucinations, 15,758 relationship hallucinations, and 2,904 scene hallucinations.

HalLoc-Instruct This subset contains 60,862 samples. The source text is drawn from sentence claims in the GQA dataset. Additionally, GPT-4 generates prompts tailored to each sentence to model diverse instructions. The average number of words per sample is 7.21, with an average of two hallucinated words (28%). It includes 15,000 object hallucinations, 17,306 attribute hallucinations, 19,402 relationship hallucinations, and 4,084 scene hallucinations.

HalLoc-Caption This subset contains 39,237 samples. The source text originates from the Stanford Image Para-

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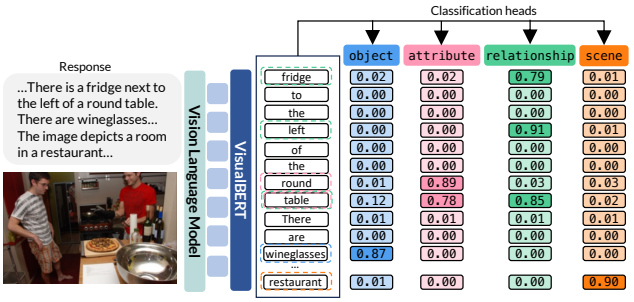


Figure 4. *HalLocalizer* employs a bidirectional VisualBERT encoder and 4 linear classification heads for token-level localization of each four hallucination types.

graphs [15] and Localized Narratives [24] datasets. It is the longest subset, with an average of 57.53 words per sample and 2.72 hallucinated words (5%). It is also the most diverse in terms of the number of injected hallucinations per source text, ranging from 0 to 6. It includes 24,037 object hallucinations, 17,608 attribute hallucinations, 9,192 relationship hallucinations, and 3,136 scene hallucinations. The relative imbalance between object, attribute, and relationship hallucinations compared to other subsets arises from higher failure rates during hallucination injection for relationship types.

5. *HalLocalizer*

The goal of *HalLoc* is to support the development of plug-and-play hallucination detection modules that can be seamlessly integrated into existing Vision-Language Models (VLMs). As an initial exploration in this direction, we introduce ***HalLocalizer***, a family of lightweight token-level hallucination detection models trained on *HalLoc*.

5.1. Task

The task of *HalLocalizer* is to identify the token-level location and type of hallucinations in arbitrary model responses for a given image. Each token is evaluated to determine whether it constitutes a hallucination for each type. Formally, let $H(t)$ represent the label for token t, where $H(t) \in \{0, 1\}$ indicates whether token t is a hallucination (1) or not (0). For each type h, a localization model predicts a function $H_h(t)$, which determines whether a given token t is classified as a hallucination of type h, $h \in \{\text{object}, \text{attribute}, \text{relationship}, \text{scene}\}$.

5.2. Architecture

HalLocalizer features two primary architectures designed to support immediate detection of token-level hallucinations, including confidence levels, during response generation.

The first architecture leverages the vision-language model embeddings—specifically, the last hidden state of the vision-language model during response generation—as input to the detection module. The second architecture directly utilizes the text response generated by the vision-language model as input to the detection module. In both cases, the detection module consists of a VisualBERT [17] encoder followed by four linear classification heads, one for each type of hallucination.

An illustration of our proposed architecture is provided in Figure 4. When *HalLocalizer* is integrated into existing vision-language models, it enables them to generate responses alongside token-level hallucination probabilities for each hallucination type.

5.3. Training Details

In both architectures, only the detection modules are trainable—the parameters of the VisualBERT encoder and the four linear classification heads. For models that utilize LVLM embeddings, we cache these embeddings prior to training to improve efficiency. We set the sequence length to 512 in all experiments and fixed the batch size at 64. We train *HalLocalizer* using the AdamW optimizer [21] with $\beta = (0.9, 0.999)$, $\epsilon = 1.0 \times 10^{-8}$, and a weight decay of 1.0×10^{-2}. Training is conducted for 25 epochs on four A6000 GPUs, resulting in an average training time of 10 hours. We employ a learning rate of 1.0×10^{-6}, with a cosine annealing scheduler applied across all trials.

6. Experiments

We evaluate the performance of *HalLocalizer* on the *HalLoc* dataset on each of the three subsets: VQA, Instruct, and Caption. The methods compared[1] include a naive baseline classifier that always predicts 1 (*Always 1*), CHAIR [26] (*CHAIR*)[2], and our proposed HalLocalizer (*HalLocalizer*) with various Large Vision-Language Model (LVLM) embeddings—MiniGPT4 [37], InstructBLIP [6], LLaVA [20], InternVL [5]—as well as without any LVLM embedding. The evaluation metrics[3] are Precision (*P*), Recall (*R*), and F1-score (*F1*) computed across four categories: Object, Attribute, Relationship, and Scene (see Table 4).

6.1. VQA

HalLocalizer demonstrates superior performance, particularly when combined with LVLM embeddings. HalLocalizer with InternVL embedding attains the highest F1-scores across all categories: 0.71 for Object, 0.94 for Attribute, 0.71 for Relationship, and 0.93 for Scene.

[1] Due to the inability of other detection methods to perform token-level hallucination detection, we cannot directly compare the performance.

[2] CHAIR is limited to detecting object hallucinations only.

[3] Each classification head's threshold is individually set to its optimal value based on the validation set for each subset.

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Table 4. Token-level hallucination detection results on HalLoc. We compare HalLocalizer with existing methods across three tasks: VQA, Instruction Following, and Image Captioning. For each hallucination type, we report Precision (P), Recall (R), and F1-score (F1). CHAIR and Always-1 are baseline methods.

Subset	Method	LVLM Embedding	Object			Attribute			Relationship			Scene		
			P	R	F1	P	R	F1	P	R	F1	P	R	F1
VQA	CHAIR	-	0.27	0.15	0.19	-	-	-	-	-	-	-	-	-
	Always 1	-	0.28	1.00	0.44	0.28	1.00	0.44	0.28	1.00	0.44	0.06	1.00	0.12
	HalLocalizer	MiniGPT4	0.54	0.87	0.67	0.83	0.91	0.87	0.53	0.82	0.64	0.86	0.75	0.80
		InstructBLIP	0.54	0.74	0.62	0.63	0.78	0.69	0.42	0.83	0.56	0.9	0.80	0.84
		LLaVA	0.57	0.87	0.69	0.87	0.95	0.90	0.54	0.83	0.66	0.94	0.85	0.89
		InternVL	0.61	0.87	**0.71**	0.91	0.96	**0.94**	0.60	0.86	**0.71**	0.94	0.93	**0.93**
		w/o Embedding	0.59	0.82	0.69	0.71	0.83	0.77	0.48	0.77	0.60	0.78	0.84	0.81
Instruct	CHAIR	-	0.15	0.12	0.13	-	-	-	-	-	-	-	-	-
	Always 1	-	0.04	1.00	0.07	0.08	1.00	0.15	0.14	1.00	0.26	0.01	1.00	0.02
	HalLocalizer	MiniGPT4	0.89	0.68	0.77	0.90	0.95	0.92	0.68	0.81	0.74	0.91	0.96	0.93
		InstructBLIP	0.88	0.70	0.78	0.91	0.95	0.93	0.73	0.90	0.81	0.91	0.95	0.93
		LLaVA	0.87	0.72	0.79	0.93	0.95	0.94	0.75	0.89	0.81	0.96	0.93	**0.94**
		InternVL	0.89	0.71	0.79	0.94	0.96	0.95	0.75	0.95	**0.84**	0.94	0.93	**0.94**
		w/o Embedding	0.94	0.73	**0.82**	0.94	0.99	**0.97**	0.75	0.92	**0.83**	0.93	0.94	**0.94**
Caption	CHAIR	-	0.03	0.08	0.04	-	-	-	-	-	-	-	-	-
	Always 1	-	0.01	1.00	0.02	0.02	1.00	0.02	0.01	1.00	0.01	0.002	1.00	0.005
	HalLocalizer	MiniGPT4	0.26	0.27	0.27	0.15	0.19	0.17	0.33	0.29	0.31	0.58	0.35	0.43
		InstructBLIP	0.37	0.38	0.37	0.26	0.29	0.28	0.52	0.44	0.47	0.62	0.51	0.56
		LLaVA	0.47	0.40	0.43	0.31	0.27	0.29	0.56	0.39	0.46	0.73	0.48	0.58
		InternVL	0.56	0.60	0.58	0.42	0.34	0.37	0.52	0.41	0.46	0.26	0.25	0.25
		w/o Embedding	0.66	0.70	**0.68**	0.67	0.60	**0.64**	0.71	0.71	**0.71**	0.85	0.69	**0.76**

6.2. Instruct

Remarkably, HalLocalizer without any LVLM embeddings achieves the highest F1-scores for Object (0.82) and Attribute (0.97), and ties for the best F1-score in Scene (0.94). For the Relationship category, both HalLocalizer with InternVL and without embedding perform competitively, with F1-scores of 0.84 and 0.83, respectively.

6.3. Caption

HalLocalizer without any LVLM embedding outperforms all other configurations, achieving F1-scores of 0.68 for Object, 0.64 for Attribute, 0.71 for Relationship, and 0.76 for Scene.

7. Analysis

7.1. Results of Human Evaluation

For human evaluation, we employ six AMT workers to assess hallucinated contents in HalLoc. We randomly sample 100 data points from each of the sub-benchmarks. For each benchmark type, workers are asked to rate each sample on two binary (yes/no) questions:

1. **Generation Accuracy**: Are the hallucinated tokens *truly* hallucinated? (i.e., Do the hallucinated tokens convey false information in the context of the image?)
2. **Injection Accuracy**: Are the hallucinated segments adequately injected into the source text in terms of context?

Table 5. Average accuracies across *HalLoc*. Workers assigned to each benchmark show unanimous agreement on their ratings.

	Generation	Injection
Accuracy	0.91	0.98
Worker Agreement	100 %	100 %

We report the average accuracies across HalLoc in Table 5. We ensure that AMT workers receive fair wages of approximately $18 per hour.

7.2. Limitations of Using Token Log Probabilities

Log probabilities from VLMs can offer a simple alternative to lightweight hallucination detection modules. However, they come with notable limitations: First, log probabilities cannot distinguish between different types of hallucinations. Second, we empirically tested the effectiveness of token-level log probabilities in identifying hallucinated tokens using the HalLoc dataset. The results are summarized in Table 6. Although a high log probability may numerically

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Table 6. Comparison of HalLocalizer and token-level log probabilitiy baselines across HalLoc. Total stands for binary hallucination labels, encompassing all hallucination types.

HalLoc Subset	Method	Model Arch	Total		
			P	R	F1
VQA	HalLocalizer	InternVL + VisualBERT	0.91	1	0.95
	LogProb	LLaVA	0.9	1	0.95
Instruct	HalLocalizer	VisualBERT	0.86	0.97	0.91
	LogProb	LLaVA	0.32	0.94	0.47
Caption	HalLocalizer	VisualBERT	0.72	0.71	0.71
	LogProb	LLaVA	0.12	0.30	0.17

suggest confidence, without proper calibration, this confidence is not necessarily justified. Calibrated uncertainty from estimates from HalLocalizer adjusts these probabilities to better align with real-world correctness.

In the VQA subset, both methods perform comparably well, each achieving a high F1-score of 0.95, indicating strong capabilities in detecting hallucinations within question-answering contexts. However, in the Instruct subset, HalLocalizer significantly outperforms the LogProb method, achieving F1-scores of 0.91 and 0.47, respectively. This disparity is primarily attributed to the LogProb method's low precision of 0.32, suggesting that it produces a substantial number of false positives despite the high recall of 0.94.

In the Caption subset, the performance gap widens further: *HalLocalizer* achieves an F1-score of 0.71, whereas the LogProb method yields a considerably lower F1-score of 0.17. These results demonstrate that *HalLocalizer* consistently delivers more reliable hallucination detection across diverse tasks, particularly excelling in scenarios where the LogProb method struggles due to its inability to handle longer, more complex responses accurately.

7.3. Ablation Study on LVLM Embeddings

To thoroughly examine the impact of using LVLM embeddings compared to textual inputs, we investigate three variants of HalLocalizer. We employ InternVL as the LVLM in these experiments, given its robust performance among models using LVLM embeddings. The first variant, *VisualBERT*, processes the text responses generated by the vision-language model as input to the detection module and utilizes only its own vision encoder for visual information. The second variant, *LVLM*, is specifically crafted for this study and replaces VisualBERT with a BERT encoder, relying exclusively on LVLM embeddings for visual input. The third variant, *LVLM+VisualBERT*, feeds the embeddings generated by the vision-language model into the detection module, integrating both the LVLM embeddings and the VisualBERT vision encoder.

Table 7. Ablation study on LVLM Embeddings

HalLoc Subset	Method	Obj	Attr	Rel	Scen
		F1	F1	F1	F1
VQA	VisualBERT	0.69	0.77	0.60	0.81
	LVLM	0.72	0.95	0.70	0.93
	LVLM + VisualBERT	0.71	0.94	0.71	0.93
Instruct	VisualBERT	0.82	0.97	0.83	0.94
	LVLM	0.68	0.95	0.84	0.95
	LVLM + VisualBERT	0.79	0.95	0.84	0.94
Caption	VisualBERT	0.68	0.64	0.71	0.71
	LVLM	0.58	0.41	0.54	0.65
	LVLM + VisualBERT	0.58	0.37	0.46	0.25

As shown in Table 7, in the VQA subset, both *LVLM* and *LVLM+VisualBERT* outperform *VisualBERT*, particularly in recognizing attributes and scenes, achieving F1-scores of up to 0.95 and 0.93, respectively. In contrast, in the Instruct and Caption subsets, *VisualBERT* surpasses the other models, suggesting that LVLM embeddings may not provide additional benefits in hallucination detection for more extended responses. This observation may be related to a decrease in calibration as the response length increases, causing LVLM embeddings to convey uncertain information less reliably. However, definitive conclusions cannot be drawn at this stage and we leave this topic for future research.

8. Conclusion

In this paper, we introduced HalLoc, a large-scale dataset comprising 155K samples across visual question answering, instruction-following, and image captioning tasks. Each sample is annotated at the token level with specific types of hallucination, allowing fine-grained training and evaluation. Using HalLoc, we developed baseline models capable of concurrent hallucination detection with minimal overhead. Unlike existing binary approaches, HalLocalizer provides probabilistic assessments, capturing confidence levels for ambiguous cases. This lightweight module can be seamlessly integrated into existing Vision-Language Models (VLMs), enhancing their reliability without compromising efficiency.

Our work lays the foundation for more advanced and lightweight hallucination detection modules, which also opens new directions for future research on sophisticated hallucination mitigation strategies [14, 36]. This scalable approach strikes a balance between accuracy and efficiency, advancing research in the practical deployment of trustworthy VLMs.

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